

Here are the answers to the **Stage 9 Lower Secondary Science Worksheet: Covalent, Ionic, and Giant Structures**:

Part A: Knowledge Retrieval

1. Fill in the Blanks

1. An **ionic** bond is formed when metal atoms lose electrons and non-metal atoms gain electrons, resulting in the **transfer** of electrons.
2. A **covalent** bond is formed when two non-metal atoms interact by the **sharing** of pairs of electrons.
3. Substances like water (H_2O) and carbon dioxide (CO_2) consist of small groups of atoms called **molecules**. They have a **simple** molecular structure.
4. Sodium chloride forms a **giant** ionic lattice with **high** melting and boiling points.
5. Diamond and graphite are examples of **giant** covalent structures.

2. True or False

1. **F — Correction:** molten (or dissolved in water / in liquid state). (Ionic compounds cannot conduct electricity when solid because ions are fixed in position).
2. **F — Correction:** between. (Simple covalent molecules have strong covalent bonds inside the molecule, but weak forces between molecules).
3. **T**

Part B: Categorization & Matching

3. The Sorting Table

Giant Ionic Lattice	Simple Covalent Molecule	Giant Covalent Structure
Magnesium oxide (MgO)	Carbon dioxide (CO_2)	Diamond (C)
Sodium chloride ($NaCl$)	Water (H_2O)	Silicon dioxide (SiO_2)

4. Match the Property to the Structure

- **Giant Ionic Structure → Physical Property:** High melting point; conducts electricity only when molten or dissolved in water. → **Explanation:** Ions are free to move and carry an electric charge when liquid or in solution.
- **Simple Covalent Structure → Physical Property:** Low melting point; does not conduct electricity. → **Explanation:** Weak intermolecular forces between molecules take little energy to overcome.
- **Giant Covalent Structure → Physical Property:** Very high melting point; does not conduct electricity (except graphite). → **Explanation:** Strong covalent bonds extend throughout the entire lattice network.

Part C: Bonding Diagrams & Visuals

5. Covalent Dot-and-Cross Diagrams

a) Water (H_2O)

- **Oxygen (O):** Central atom with 6 outer shell electrons.
- **Hydrogen (H):** 2 hydrogen atoms, each sharing 1 electron with Oxygen.
- **Diagram Description:**
 - Oxygen has **2 single covalent bonding pairs** (one with each Hydrogen).
 - Oxygen retains **2 lone pairs** (4 unshared outer electrons).
 - Each Hydrogen has a full shell of 2 electrons; Oxygen has a full outer shell of 8 electrons.

b) Methane (CH_4)

- **Carbon (C):** Central atom with 4 outer shell electrons.
- **Hydrogen (H):** 4 hydrogen atoms surrounding Carbon.
- **Diagram Description:**
 - Carbon forms **4 single covalent bonding pairs** (1 shared pair per $C-H$ bond).
 - No lone pairs on the central Carbon atom.
 - Carbon achieves a full outer shell of 8 electrons; each Hydrogen has 2 electrons.

6. Ionic Dot-and-Cross Diagrams

Sodium Chloride ($NaCl$)

- **Sodium ion (Na^+):**
 - Sodium atom loses its single outer electron to form a $[Na]^+$ ion.
 - Draw square brackets around the atom symbol: $[Na]^+$, showing a full shell underneath (or empty valence shell).
- **Chloride ion (Cl^-):**
 - Chlorine atom gains 1 electron to complete its outer shell of 8.
 - Draw square brackets around the atom symbol: $[Cl]^-$ with 7 dots (or crosses) representing Chlorine's original electrons and 1 cross (or dot) representing the transferred electron from Sodium.

Part D: Cambridge Checkpoint & Workbook Style

Questions

Question 7

(a)

- **Structure Type:** Giant Ionic Lattice (or Giant Ionic Structure).
- **Reason:** It has a high melting point and only conducts electricity when molten or dissolved, not as a solid.

(b)

- **Substance:** Substance Y.
- **Reason:** Carbon dioxide is a simple covalent molecule. It has a low melting point due to weak intermolecular forces and does not conduct electricity in any state.

(c)

- Diamond is a giant covalent structure where every carbon atom is strongly bonded to 4 other carbon atoms in a 3D lattice network. Breaking these many strong covalent bonds requires a massive amount of heat energy.

Question 8

(a)

- **Structure Classification:** Giant Covalent Structures (or Macromolecular Structures).

(b)

- **Graphite (Structure B):** Consists of flat layers held together by weak forces. These layers can easily slide over each other, making it slippery and useful as a lubricant.
- **Diamond (Structure A):** Has a rigid 3D tetrahedral network where each carbon atom is strongly bonded to 4 others. This rigid, locked structure makes diamond extremely hard and suitable for heavy-duty cutting tools.

(c)

- **Graphite (Structure B)** has delocalized electrons (each carbon atom only uses 3 of its 4 outer electrons in bonding, leaving 1 free electron per atom) that are free to move throughout the structure and carry charge.
- **Diamond (Structure A)** uses all 4 outer electrons in strong covalent bonds, so there are no free (delocalized) electrons to carry an electric charge.